

Date : 16-02-2026

Problem 1: Self Introduction

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Self Introduction </title>
</head>
<body>
  <header>
    <h1> Self Introduction</h1>
    <p> Good afternoon My name is Dharani. I completed my Graduation in ECE. My technical skills are Python,
CoreJava in addition to this i am good at Communication </p>
  </header>
  <hr>
  <section>
    <h2> Hobbies </h2>
    <ul style="list-style-type: square;">
      <li> Pencil Drawing</li>
      <li> Cooking </li>
      <li> Singing </li>
      <li> Diy Crafts </li>
    </ul>
  </section>
  <hr>
  <section>
    <h2> Daily routine </h2>
    <ol type="A">
      <li> Wake Up </li>
      <li> Ready </li>
      <li> Breakfast</li>
      <li> do daily tasks </li>
      <li> Self Development </li>
      <li> Lunch </li>
      <li> Works </li>
      <li> Sleep </li>
    </ol>
  </section>
  <hr>
  <section>
    <h2> Student Marks </h2>
    <table border="1" cellpadding="8">
      <tr>
        <th> Student Name </th>
        <th> Subject </th>
        <th> Marks </th>
      </tr>
      <tr>
        <td> Sony </td>
        <td> Communication </td>
        <td> 90 </td>
      </tr>
      <tr>
        <td> Riya</td>
        <td> DBMS </td>
        <td> 95 </td>
      </tr>
      <tr>
        <td> Divya </td>
        <td> Digital Electronics </td>
        <td> 87 </td>
      </tr>
    </table>
  </section>
</body>
</html>
```

```
</tr>
</table>
</section>
</body>
</html>
```

Problem 2: Restaurant Menu

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Restaurant</title>
</head>
<body>
  <header> <h1 style="text-align:center" >Andhra's Kitchen </h1></header>
  <p>Andhra Kitchen is a place where authentic Telugu traditions come alive through food.
We serve tasty and tummy filling dishes made with homemade masalas and natural ingredients.
Our recipes bring the true essence of Andhra flavors to every plate.
Don't miss our special Mirchi Masala Rice, prepared with our own unique recipe.
Visit us and experience the real touch of Andhra cuisine. </p>
```

```
  <section>
    <h2> <i>Menu Categories </i> </h2>
    <ul>
      <li>Main Course</li>
      <li>Breakfast</li>
      <li>Beverages</li>
    </ul>
  </section>
  <section>
    <h2> <u>Price List </u></h2>
    <table border="1" cellpadding="10">
      <tr>
        <th> Item Name </th>
        <th> Category </th>
        <th> Price ($) </th>
      </tr>
      <tr>
        <td> Upma </td>
        <td> Breakfast </td>
        <td> 90 </td>
      </tr>
      <tr>
        <td> Gongura Pachadi </td>
        <td> Main Course </td>
        <td> 60 </td>
      </tr>
      <tr>
        <td> Andhra Chicken Curry </td>
        <td> Main Course </td>
        <td> 200 </td>
      </tr>
      <tr>
        <td> Lassi </td>
        <td> Beverages </td>
        <td> 100 </td>
      </tr>
    </table>
  </section>
</body>
</html>
```

Problem 3: Grocery Checklist

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Weekly Grocery Checklist</title>
</head>
<body>
  <h2>Weekly Grocery Checklist</h2>
  <section>
    <h2> high-priority grocery items </h2>
    <ol>
      <li>Rice</li>
      <li>Milk</li>
      <li>Vegetables</li>
      <li>Cooking Oil</li>
    </ol>
  </section>
  <section>
    <h2> non-essential items </h2>
    <ul style="list-style-type:square;">
      <li>Snacks</li>
      <li>Ice cream</li>
      <li>Soft drinks</li>
    </ul>
  </section>
</body>
</html>
```

Problem 4: About Me

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>About Me Page</title>
</head>
<body>
  <header>
    <h1 style="text-align: center;"> Golla Dharani </h1>
    <p>Good afternoon My name is Dharani. I completed my Graduation in ECE.My technicals skills are <b>Python,
CoreJava</b> in addition to this <mark>i am good at Communication</mark> </p>
  </header>
  <section>
    <h2> My Goals </h2>
    <ul>
      <li>To get a good job in the IT industry.</li>
      <li> To keep learning new technologies and upgrading my knowledge.</li>
      <li> To maintain a balanced life with personal growth and career success.</li>
      <li> To adapt quickly to a professional work environment and deliver quality results</li>
      <li> To begin my career as a fresher and grow step by step in a good organization</li>
    </ul>
  </section>
  <footer>
    <p> <b> <i>Created by Dharani - 2026</i></b> </p>
  </footer>
</body>
</html>
```

Problem 5: Student Marks Report

```
<!DOCTYPE html>
```

```

<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>student_marks.html</title>
</head>
<body>
  <header>
    <h1 align = "center"><u>Student Marks Report</u></h1>
  </header>
  <section>
    <h2> Students Python Marks:</h2>
    <table border="2" cellpadding="8">
      <tr>
        <th>Roll No</th>
        <th>Name</th>
        <th>Subject</th>
        <th>Total Marks</th>
        <th>Marks Obtained</th>
        <th>Result</th>
      </tr>
      <tr>
        <td>101</td>
        <td>Sony</td>
        <td>Python</td>
        <td>60</td>
        <td>58</td>
        <td style="color:green;">Pass</td>
      </tr>
      <tr>
        <td>102</td>
        <td>meena</td>
        <td>Python</td>
        <td>60</td>
        <td>56</td>
        <td style="color:green;">Pass</td>
      </tr>
      <tr>
        <td>103</td>
        <td>Siri</td>
        <td>Python</td>
        <td>60</td>
        <td>49</td>
        <td style="color:green;">Pass</td>
      </tr>
      <tr>
        <td>104</td>
        <td>Riya</td>
        <td>Python</td>
        <td>60</td>
        <td>60</td>
        <td style="color:green;">Pass</td>
      </tr>
      <tr>
        <td>105</td>
        <td>Rishika</td>
        <td>Python</td>
        <td>60</td>
        <td>20</td>
        <td style="color:red;">Fail</td>
      </tr>
    </table>
  </section>

```



```
</section>
<hr>
<section>
  <h2> Top subjects </h2>
  <ol>
    <li>Python</li>
    <li>AWS Basics</li>
    <li>ML</li>
    <li>SQL</li>
    <li>CyberSecurity</li>
  </ol>
</section>
</body>
</html>
```

Problem 6: Restaurant Menu

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>restaurant_menu.html</title>
</head>
<body>
  <header>
    <h1>India's Kitchen</h1>
    <p>Taste the Tradition of India in Every Bit</p>
  </header>
  <nav>
    <a href="#home">Home</a>
    <a href="#menu">Menu</a>
    <a href="#">Contact</a> <!-- written in footer -->
  </nav>
  <section id = "home" style="text-align:center; padding:20;">
    <p>
      Welcome, India's Kitchen brings you authentic Indian food made with love and tradition.
      We use fresh ingredients and homemade spices to give you rich and delicious flavors.
      Enjoy home-style meals that are healthy, tasty, and satisfying.
    </p>
  </section>
  <section id="menu">
    <h2>Explore Our Menu – A Taste of India Awaits!</h2>
    <table border="1" cellpadding="9">
      <tr>
        <th>Item Name</th>
        <th> Category</th>
        <th> Price</th>
        <th> Availability </th>
      </tr>
      <tr>
        <td>Samosa</td>
        <td>Starter</td>
        <td>40</td>
        <td>Available</td>
      </tr>
      <tr>
        <td>Paneer Tikka</td>
        <td>Starter</td>
        <td>160</td>
        <td>Available</td>
      </tr>
```

```

<tr>
  <td>Veg Biryani</td>
  <td>Main Course</td>
  <td>180</td>
  <td>Available</td>
</tr>

<tr>
  <td>Paneer Butter Masala</td>
  <td>Main Course</td>
  <td>220</td>
  <td>Available</td>
</tr>

<tr>
  <td>Masala Dosa</td>
  <td>Breakfast</td>
  <td>90</td>
  <td>Available</td>
</tr>

<tr>
  <td>Gulab Jamun</td>
  <td>Dessert</td>
  <td>80</td>
  <td>Available</td>
</tr>

<tr>
  <td>Pootharekulu</td>
  <td>Dessert</td>
  <td>100</td>
  <td>Not Available</td>
</tr>

<tr>
  <td>Masala Chai</td>
  <td>Drinks</td>
  <td>30</td>
  <td>Available</td>
</tr>
</table>

</section>
<section>
  <h2 style="color:green;">Top 5 Recommended Items</h2>
  <ol>
    <li>Paneer Butter Masala</li>
    <li>Veg Biryani</li>
    <li>Chicken 65</li>
    <li>Masala Dosa</li>
    <li>Gulab Jamun</li>
  </ol>
</section>
<section>
  <h2>Available Offers</h2>
<ul>
  <li>10% discount on all Main Course items</li>
  <li>Buy 1 Get 1 Free on selected Starters</li>
  <li>Free dessert with orders above ₹500</li>
  <li>Special lunch combo at discounted price</li>
  <li>Free soft drink with every family meal</li>
</ul>

```

<footer>
<p>Thank you for visiting!</p>
<p>Email:
IndiaKitchen@gmail.com</p>
</footer>

</body>
</html>

Day2:1st part Date:17-01-2026

Problem1: navigation + registration form

Home.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Home</title>
  <style>
    nav{
      background-color: blueviolet;
      padding: 10px;
      width: 90%;
    }
    nav a{
      padding: 10px;
      margin: 15px;
      color: white;
      text-decoration: none;
    }
    nav a:hover{
      background-color: brown;
    }

    form{
      width: 50%;
      margin: auto;
      padding: 15px;
      border: 1px solid gray;
      border-radius: 10px;
    }

    label{
      display: block;
      margin-top: 10px;
    }

    input, select{
      width: 90%;
      margin-top: 10px;
      padding: 10px;
      border: 1px solid black;
      border-radius: 10px;
    }

    .gender{
      width: auto;
      margin-right: 5px;
    }

    button{
      width: 90%;
      background-color: yellow;
      margin: 10px;
      padding: 10px;
      border: 2px solid black;
      cursor: pointer;
    }
  </style>
</head>
<body>

  <nav>
    <a href="home.html">Home</a>
    <a href="contact.html">Contact</a>
  </nav>

  <form>
```

```

<label>Name:</label>
<input type="text" name="name" placeholder="Enter your name" required>

<label>Email:</label>
<input type="email" name="email" placeholder="Enter your email" required>

<label>Password:</label>
<input type="password" name="password" placeholder="Enter your password" required>

<label>Gender:</label>
<input type="radio" name="gender" value="Female" class="gender" required> Female
<input type="radio" name="gender" value="Male" class="gender"> Male

<label>Course Selection:</label>
<select name="course" required>
  <option value="">Select Course</option>
  <option value="Python">Python</option>
  <option value="Java">Java</option>
  <option value="C">C</option>
</select>

<button type="submit">Send Message</button>
</form>

```

```

</body>
</html>
Contact.html:
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Contact</title>
  <style>
    nav{
      background-color: blueviolet;
      padding: 10px;
      width: 90%;
    }
    nav a{
      padding: 10px;
      margin: 15px;
      color: white;
      text-decoration: none;
    }
    nav a:hover{
      background-color: brown;
    }

    body{
      font-family: Arial, sans-serif;
    }

    footer{
      margin-top: 20px;
    }
  </style>
</head>
<body>

  <nav>
    <a href="home.html">Home</a>
    <a href="contact.html">Contact</a>
  </nav>

  <h2>Contact Us</h2>

  <footer>
    <p>
      Email us:
      <a href="mailto:doubt12@email.com">doubt12@email.com</a>
    </p>

```

```

        <p>
            Phone:
            <a href="tel:99999999">99999999</a>
        </p>
    </footer>

</body>
</html>
*****
Problem2: Simple Website Navigation System
index.html:
<!DOCTYPE html>
<html lang="en">
<head>
    <title>Coaching Center</title>
</head>
<body>

<nav>
    <a href="index.html">Home</a> |
    <a href="courses.html">Courses</a> |
    <a href="contact.html">Contact</a>
</nav>

<p>
<i>
Smart Coaching Center offers expert guidance and quality education to help students succeed and build a strong future.
</i>
</p>

</body>
</html>
courses.html:
<!DOCTYPE html>
<html lang="en">
<head>
    <title>Coaching Center - Courses</title>
</head>
<body>

<nav>
    <a href="index.html">Home</a> |
    <a href="courses.html">Courses</a> |
    <a href="contact.html">Contact</a>
</nav>

<section id="programming">
    <p>Explore our programming courses</p>
</section>

<h2>Explore Our Courses</h2>

<h3>Programming Languages</h3>
<ul>
    <li>Python</li>
    <li>Java</li>
    <li>C</li>
    <li>C++</li>
    <li>C#</li>
    <li>JavaScript</li>
    <li>Go (Golang)</li>
</ul>

<h3>Cloud Courses</h3>

```

```
<ul>
  <li>AWS</li>
  <li>Azure</li>
  <li>Google Cloud Platform (GCP)</li>
  <li>Cloud Computing Basics</li>
  <li>DevOps on Cloud</li>
  <li>Docker</li>
  <li>Kubernetes</li>
  <li>Terraform</li>
  <li>Cloud Security</li>
  <li>Serverless Computing</li>
</ul>
```

<h3>Testing Courses</h3>

```
<ul>
  <li>Selenium</li>
  <li>Playwright</li>
  <li>Cypress</li>
  <li>JUnit</li>
  <li>Manual Testing</li>
  <li>API Testing (Postman)</li>
  <li>Mobile Testing (Appium)</li>
  <li>Performance Testing (JMeter)</li>
  <li>TestNG</li>
</ul>
```

<h2>Internal Link</h2>

Go to Programming Section

```
</body>
</html>
contact.html:
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Coaching Center - Contact</title>
</head>
<body>
```

```
<nav>
  <a href="index.html">Home</a> |
  <a href="courses.html">Courses</a> |
  <a href="contact.html">Contact</a>
</nav>
```

```
<p>
For any enquiries about admissions, courses, or schedules, please reach out to us.
</p>
```

```
<footer>
  <p>
    Email us:
    <a href="mailto:doubt12@email.com">doubt12@email.com</a>
  </p>
```

```
<p>
  Phone:
  <a href="tel:99999999">99999999</a>
```

```
</p>

<p>
  Follow us on:
  <a href="https://www.facebook.com" target="_blank">Facebook</a>
</p>
</footer>

</body>
</html>
*****
```

Problem3: Student Registration Form with Input Types (Level-1)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Student Registration Form</title>
</head>
<body>

<h2>Student Registration Form</h2>

<form>

  <label for="name">Name:</label>
  <input type="text" id="name" name="name" required>

  <br><br>

  <label for="email">Email:</label>
  <input type="email" id="email" name="email" required>

  <br><br>

  <label for="phone">Phone Number:</label>
  <input type="tel" id="phone" name="phone"
    pattern="[0-9]{10}"
    placeholder="Enter 10 digit number"
    required>

  <br><br>

  <label for="age">Age:</label>
  <input type="number" id="age" name="age"
    min="18" max="60"
    required>

  <br><br>

  <label for="dob">Date of Birth:</label>
  <input type="date" id="dob" name="dob" required>

  <br><br>

  <button type="submit">Submit</button>

</form>

</body>
```


</html>

Problem4: E-Commerce Product Website

Index.html

<!DOCTYPE html>

<html>

<head>

<title>Online Store</title>

<!-- Internal CSS -->

<style>

body {

font-family: Arial, sans-serif;

margin: 0;

padding: 0;

background-color: #f4f4f4;

}

/* Navigation Bar */

nav {

background-color: #333;

padding: 15px;

text-align: center;

}

nav a {

color: white;

text-decoration: none;

margin: 0 20px;

font-size: 18px;

font-weight: bold;

}

nav a:hover {

color: #ff9800;

}

/* Sections */

section {

padding: 40px;

background: white;

margin: 20px;

border-radius: 8px;

box-shadow: 0 0 10px rgba(0,0,0,0.1);

}

h2 {

color: #333;

}

/* Category List */

ul {

list-style-type: square;

padding-left: 20px;

}

ul li {

margin: 10px 0;

```

        font-size: 18px;
    }

    ul li a {
        text-decoration: none;
        color: blue;
        font-weight: bold;
    }

    ul li a:hover {
        color: red;
    }

    /* Offers Ordered List */
    ol {
        padding-left: 20px;
    }

    ol li {
        margin: 10px 0;
        font-size: 18px;
        color: #444;
    }

    /* Anchor Links */
    .anchor-links {
        text-align: center;
        margin: 20px 0;
    }

    .anchor-links a {
        text-decoration: none;
        margin: 0 15px;
        font-weight: bold;
        color: green;
    }

    .anchor-links a:hover {
        color: darkgreen;
    }
</style>
</head>

<body>

<!-- Navigation Menu -->
<nav>
    <a href="#">Home</a>
    <a href="electronics.html">Electronics</a>
    <a href="#">Checkout</a>
</nav>

<!-- Internal Anchor Links -->
<div class="anchor-links">
    <a href="#categories">Go to Categories</a>
    <a href="#offers">Go to Offers</a>
</div>

```

```
<!-- Categories Section -->
<section id="categories">
  <h2>Categories</h2>
  <ul>
    <li><a href="electronics.html">Electronics</a></li>
    <li>Fashion</li>
    <li>Grocery</li>
    <li>Sports</li>
  </ul>
</section>

<!-- Offers Section -->
<section id="offers">
  <h2>Special Offers</h2>
  <ol>
    <li>Up to 50% off on Smartphones</li>
    <li>Buy 1 Get 1 Free on Headphones</li>
    <li>30% Discount on Fashion Items</li>
    <li>20% Cashback on Grocery Orders</li>
    <li>Flat 40% Off on Sports Equipment</li>
  </ol>
</section>

</body>
</html>
electronics.html:
<!DOCTYPE html>
<html>
<head>
  <title>Electronics</title>

  <!-- Internal CSS -->
  <style>
    body {
      font-family: Arial, sans-serif;
      margin: 0;
      padding: 0;
      background-color: #f4f4f4;
    }

    /* Breadcrumb */
    .breadcrumb {
      background-color: #eee;
      padding: 10px 20px;
      font-size: 16px;
    }

    .breadcrumb a {
      text-decoration: none;
      color: blue;
      font-weight: bold;
    }

    .breadcrumb a:hover {
      color: red;
    }

    /* Section Styling */
```

```
section {
  padding: 30px;
  margin: 20px;
  background: white;
  border-radius: 8px;
  box-shadow: 0 0 10px rgba(0,0,0,0.1);
}
```

```
h2 {
  color: #333;
}
```

```
/* Table Styling */
table {
  width: 100%;
  border-collapse: collapse;
  margin-top: 20px;
}
```

```
table, th, td {
  border: 1px solid #ccc;
}
```

```
th {
  background-color: #333;
  color: white;
  padding: 10px;
}
```

```
td {
  padding: 10px;
  text-align: center;
}
```

```
tr:nth-child(even) {
  background-color: #f2f2f2;
}
```

```
/* Hierarchical List */
ul {
  list-style-type: disc;
  padding-left: 20px;
}
```

```
ul li {
  margin: 10px 0;
  font-size: 18px;
}
```

```
ul li a {
  text-decoration: none;
  color: green;
  font-weight: bold;
}
```

```
ul li a:hover {
  color: darkgreen;
}
```

```

</style>
</head>

<body>

<!-- Breadcrumb -->
<div class="breadcrumb">
  <a href="index.html">Home</a> > Electronics
</div>

<!-- Electronics Table Section -->
<section>
  <h2>Available Electronic Products</h2>

  <table>
    <tr>
      <th>Product Type</th>
      <th>Items Available</th>
      <th>Discount</th>
    </tr>
    <tr>
      <td>Mobiles</td>
      <td>25</td>
      <td>Up to 40%</td>
    </tr>
    <tr>
      <td>Laptops</td>
      <td>15</td>
      <td>Up to 30%</td>
    </tr>
    <tr>
      <td>Television</td>
      <td>10</td>
      <td>Up to 35%</td>
    </tr>
    <tr>
      <td>Headphones</td>
      <td>40</td>
      <td>Up to 50%</td>
    </tr>
  </table>
</section>

<!-- Hierarchical Linking Section -->
<section>
  <h2>Browse Electronics</h2>
  <ul>
    <li><a href="mobiles.html">Mobiles</a></li>
    <li><a href="laptops.html">Laptops</a></li>
    <li><a href="tv.html">TV</a></li>
  </ul>
</section>

</body>
</html>

```

```

Mobiles.html
<!DOCTYPE html>

```

```
<html>
<head>
  <title>Mobiles - Product Page</title>

  <style>
    body {
      font-family: Arial, sans-serif;
      margin: 0;
      padding: 0;
      background-color: #f4f4f4;
    }

    .breadcrumb {
      background-color: #eee;
      padding: 10px 20px;
      font-size: 16px;
    }

    .breadcrumb a {
      text-decoration: none;
      color: blue;
      font-weight: bold;
    }

    section {
      padding: 30px;
      margin: 20px;
      background: white;
      border-radius: 8px;
      box-shadow: 0 0 8px rgba(0,0,0,0.1);
    }

    table {
      width: 100%;
      border-collapse: collapse;
      margin-top: 20px;
    }

    table, th, td {
      border: 1px solid #ccc;
    }

    th {
      background-color: #333;
      color: white;
      padding: 10px;
    }

    td {
      padding: 10px;
      text-align: center;
    }

    tr:nth-child(even) {
      background-color: #f2f2f2;
    }

    h2 {
```

```

        margin-top: 0;
    }
</style>
</head>

<body>

<!-- Breadcrumb -->
<div class="breadcrumb">
    <a href="index.html">Home</a> >
    <a href="electronics.html">Electronics</a> >
    Mobiles
</div>

<!-- Mobiles Table -->
<section>
    <h2>Available Mobiles</h2>

    <table>
        <tr>
            <th>Mobile Name</th>
            <th>Price</th>
            <th>Stock</th>
            <th>Rating</th>
        </tr>

        <tr>
            <td>iPhone 15</td>
            <td>$999</td>
            <td>
                <meter value="70" min="0" max="100"></meter>
            </td>
            <td>4.8 ★</td>
        </tr>

        <tr>
            <td>Samsung Galaxy S23</td>
            <td>$850</td>
            <td>
                <meter value="50" min="0" max="100"></meter>
            </td>
            <td>4.6 ★</td>
        </tr>

        <tr>
            <td>OnePlus 11</td>
            <td>$720</td>
            <td>
                <meter value="40" min="0" max="100"></meter>
            </td>
            <td>4.5 ★</td>
        </tr>

        <tr>
            <td>Realme 12 Pro</td>
            <td>$450</td>
            <td>
                <meter value="80" min="0" max="100"></meter>
            </td>
            <td>4.7 ★</td>
        </tr>
    </table>
</section>

```

```

        </td>
        <td>4.3 ★ </td>
    </tr>

    <tr>
        <td>Xiaomi 13</td>
        <td>$600</td>
        <td>
            <meter value="30" min="0" max="100"></meter>
        </td>
        <td>4.4 ★ </td>
    </tr>

</table>
</section>

<!-- Order Processing Progress -->
<section>
    <h2>Order Processing</h2>
    <p>50% Processing</p>
    <progress value="50" max="100"></progress>
</section>

</body>
</html>

```

Checkout.html

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
    <title>Checkout</title>
```

```
<style>
```

```
    body {
        font-family: Arial, sans-serif;
        margin: 0;
        padding: 0;
        background-color: #f4f4f4;
    }
```

```
    /* Navigation */
```

```
    nav {
        background-color: #333;
        padding: 15px;
        text-align: center;
    }
```

```
    nav a {
        color: white;
        text-decoration: none;
        margin: 0 20px;
        font-size: 18px;
        font-weight: bold;
    }
```

```
    /* Breadcrumb */
```

```
    .breadcrumb {
        background-color: #eee;
    }
```



```

        padding: 10px 20px;
    }

    /* Form Section */
    section {
        padding: 30px;
        margin: 20px;
        background: white;
        border-radius: 8px;
        box-shadow: 0 0 8px rgba(0,0,0,0.1);
    }

    input, textarea {
        width: 100%;
        padding: 8px;
        margin: 8px 0 15px 0;
    }

    button {
        padding: 10px 20px;
        background-color: #333;
        color: white;
        border: none;
        cursor: pointer;
    }

    button:hover {
        background-color: green;
    }
</style>
</head>

<body>

<!-- Navigation Menu -->
<nav>
    <a href="index.html">Home</a>
    <a href="electronics.html" id="electronicsLink">Electronics</a>
    <a href="checkout.html">Checkout</a>
</nav>

<!-- Breadcrumb -->
<div class="breadcrumb">
    <a href="index.html">Home</a> > Checkout
</div>

<!-- Checkout Form -->
<section>
    <h2>Checkout Form</h2>

    <form>
        <!-- Full Name -->
        <label>Full Name:</label>
        <input type="text" required>

        <!-- Address -->
        <label>Address:</label>
        <textarea rows="4" required></textarea>
    </form>

```

```

<!-- Email -->
<label>Email:</label>
<input type="email" required>

<!-- Payment Method with datalist -->
<label>Payment Method:</label>
<input list="paymentMethods" required>
<datalist id="paymentMethods">
  <option value="UPI">
  <option value="Card">
  <option value="Cash">
  <option value="NetBanking">
</datalist>

<!-- Delivery Date -->
<label>Delivery Date:</label>
<input type="date" required>

<!-- Coupon Code with pattern validation -->
<label>Coupon Code:</label>
<input type="text" pattern="SHOP[0-9]{3}"
  title="Enter coupon like SHOP123">

<!-- Submit Button -->
<button type="submit">Submit Order</button>
</form>
</section>

<!-- JavaScript -->
<script>
  document.getElementById("electronicsLink")
  .addEventListener("click", function() {
    this.style.color = "green";
  });
</script>

</body>
</html>

```

Day2:Problems (day3) (CSS)

Problem1: Product Feature List Section

HTML File:

```

<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Product launch</title>

  <link rel="stylesheet" href="style.css">

</head>

```

```

<body>

  <header>

    <h1><b>PureSun Care</b></h1>

    <p>Protect your skin every day with our advanced sunscreen formula. Designed with high SPF and broad-spectrum protection, it keeps your skin safe, hydrated, and glowing under the sun</p>

  </header>

  <section class="feature">

    <h2>look over the interesting features</h2>

    <ul>

      <li><a href="#">SPF 50+ Broad Spectrum Protection</a></li>

      <li><a href="#">Lightweight & Non-Greasy</a></li>

      <li><a href="#">Water Resistant</a></li>

      <li><a href="#">Quick Absorption</a></li>

      <li><a href="#">Suitable for All Skin Types</a></li>

    </ul>

  </section>

  <footer>

    <p>Book Your Favourite product</p>

    <h3>ContantUS:</h3>

    <p>Emailus:<a href="mailto:PureSunCare@email.com">PureSunCare@email.com</a></p>

    <p>Phone:<a href="8976667886">8976667886</a></p>

  </footer>

</body>

</html>

```

CSS file:

```

*{
  margin:0;
  padding: 0;
  box-sizing: border-box;
}

header{
  justify-content:center;
  padding: 10px;
  margin:15px;
  color:blueviolet;
}

header p{
  font-family: 'Times New Roman', Times, serif;
  color: black;
}

```

```
.feature{
    font-family: 'Franklin Gothic Medium', 'Arial Narrow', Arial, sans-serif;
    font-size: medium;
    margin-left:14px;
    margin:15px 16px;
    padding: 10px ;
    color: blue;
}

.feature li{
    padding:10px;
    border: 1pxsolid black;
    border-radius:1px;
    text-decoration: none;
}

.feature a{
    text-decoration: none;
}

.feature li:hover{
    background-color: brown;
}

.feature h2{
    margin-bottom:15px;
    color: red;
}

footer{
    font-size: smaller;
    margin-left:14px;
    margin:15 16;
    padding: 10px ;
}
```

Problem=2 Responsive Blog Layout

Htmlfile

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Responsive Blog Layout</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <header class="blog-header">
    <h1><b>PureSun Care</b></h1>
    <p>Protect your skin every day with our advanced sunscreen formula. Designed with high SPF and broad-spectrum protection, it keeps your skin safe, hydrated, and glowing under the sun</p>
  </header>
<section class="blog">
```

```

<div class="card">
<h3>Why Sunscreen Is Important</h3>
<p>
    Sunscreen protects your skin from harmful UV rays...
</p>
<details>
    <summary class="read-more">Read More</summary>
    <p>
        Sunscreen protects against UVA and UVB rays,
        prevents early aging, reduces sunburn,
        and lowers the risk of skin cancer.
        Daily use keeps your skin healthy and glowing.
    </p>
</details>
</div>
<!--second card-->
<div class="card">
<h3>SPF 30 vs SPF 50</h3>
<p>
    Not sure which SPF level is right for you? Here's a quick comparison...
</p>
<details>
    <summary class="read-more">Read More</summary>

    <p>
        SPF 30 blocks about 97% of UVB rays,
        while SPF 50 blocks about 98%.
        Although the difference seems small,
        SPF 50 provides slightly better protection
        for long sun exposure and sensitive skin.
    </p>
</details>
</div>

<div class="card">
<h3>How to Apply Sunscreen Correctly</h3>
<p>
    Applying sunscreen the right way ensures maximum protection...
</p>
<details>
    <summary class="read-more">Read More</summary>

    <p>
        Apply sunscreen 15–20 minutes before going outside.
        Use a generous amount to cover all exposed skin areas.
        Don't forget your ears, neck, and hands.
        Reapply every 2 hours, especially after swimming
        or sweating, to maintain effective protection.
    </p>
</details>
</div>
</section>
</body>
</html>
CSS file:

*{
margin:0;
padding: 0;

```

```

    box-sizing: border-box;
}

header{
    justify-content:center;
    padding: 10px;
    margin:15px;
    color:blueviolet;
}
header p{
    font-family: 'Times New Roman', Times, serif;
    color: black;
}

.blog{
    display:flex;
    gap:14px;
    justify-content:center;
}

.card{
background-color: aqua;
    padding: 14px;
    margin:10px;
    border:2px solid brown;
    border-radius: 10px;
}
.card a:hover{
    background-color: rgb(239, 243, 243);
}
details {
    margin-top: 10px;
}

summary.read-more {
    cursor: pointer;
    padding: 6px 12px;
    background-color: orange;
    color: white;
    border-radius: 5px;
    list-style: none;
    display: inline-block;
}

summary.read-more:hover {
    background-color: darkorange;
}

.blog {
    display: grid;
    grid-template-columns: repeat(3, 1fr);
    gap: 20px;
}

@media(max-width:750px){
    .blog{
        flex-direction: column;
    }
}

```

```

        align-items:center;

    }
}
@media(max-width:250px){
    .blog{
        flex-direction: column;
        align-items:center;
    }
}

*****
*****

Problem3: Responsive Dashboard Layout
Htmlfile: <!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Responsive Dashboard Layout</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>

<div class="main-container">

    <!-- Sidebar -->
    <aside class="sidebar">
        <h2>Dashboard</h2>
        <ul>
            <li><a href="#">HTML</a></li>
            <li><a href="#">CSS</a></li>
            <li><a href="#">JS</a></li>
            <li><a href="#">Django</a></li>
        </ul>
    </aside>

    <!-- Main Content -->
    <section class="content">
        <div class="card">
            <h2>Text Styling</h2>
            <p>This paragraph shows font styling, colors, and spacing.</p>
        </div>

        <div class="card">
            <h2>Box Model</h2>
            <p>This card has padding, margin, and border applied.</p>
        </div>

        <div class="card">
            <h2>Responsive Design</h2>
            <p>This page becomes mobile friendly using media queries.</p>
        </div>
    </section>

</div>

</body>

```

</html>

Css file:

```
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  font-family: Arial, sans-serif;
  background-color: #f5f5f5;
}

.main-container {
  display: flex;
}

.sidebar {
  margin-top: 15px;;
  width: 250px;
  background-color: #2c3e50;
  color: white;
  padding: 20px;
  margin-bottom: 15px;
}

.sidebar h2 {
  margin-bottom: 20px;
}

.sidebar ul {
  list-style: none;
}

.sidebar ul li {
  margin: 15px 0;
}

.sidebar ul li a {
  text-decoration: none;
  color: white;
  font-weight: bold;
}

.sidebar ul li a:hover {
  color: #1abc9c;
}

.content {
  flex: 1;
  display: flex;
  flex-wrap: wrap;
  gap: 20px;
  padding: 30px;
}

.card {
```



```
background-color: white;
border: 2px solid lightgray;
padding: 20px;
width: calc(33.33% - 20px);
border-radius: 10px;
box-shadow: 2px 2px 10px rgba(0,0,0,0.1);
}
```

```
.card h2 {
  margin-bottom: 10px;
  color: darkblue;
}
```

```
.card p {
  font-size: 14px;
  line-height: 1.5;
}
```

```
@media (max-width: 992px) {
  .card {
    width: calc(50% - 20px);
  }
}
```

```
@media (max-width: 576px) {
```

```
  .main-container {
    flex-direction: column;
  }
```

```
  .sidebar {
    width: 100%;
    text-align: center;
  }
```

```
  .sidebar ul {
    display: flex;
    justify-content: center;
    gap: 20px;
  }
```

```
  .sidebar ul li {
    margin: 10px 0;
  }
```

```
  .content {
    justify-content: center;
  }
```

```
  .card {
    width: 90%;
  }
```

```
}
*****
*****
```

Problem 4: sticky navbar

Htmlfile:

<!DOCTYPE html>

```

<html>
<head>
  <title>Sticky Navbar</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

<nav class="navbar">
  <a href="#" class="active">Home</a>
  <a href="#">Shop</a>
  <a href="#">Beauty</a>
  <a href="#">Health</a>
  <a href="#">Support</a>
</nav>

<div class="content">
  <h2>Home Page</h2>
  <p>This is simple page content.</p>
  <p>Scroll down to see sticky navbar.</p>
  <p style="height:800px;"></p>
</div>

</body>
</html>

```

Css file:

```

body {
  margin: 0;
  font-family: Arial;
}

.navbar {
  background-color: black;
  position: sticky;
  top: 0;
  box-shadow: 0 2px 5px gray;
}

.navbar a {
  color: white;
  padding: 14px 20px;
  text-decoration: none;
  display: inline-block;
}

.navbar a:hover {
  background-color: green;
}

.active {
  background-color: red;
}

.content {
  padding: 20px;
}

```

Problem5: Responsive Pricing Table

<!DOCTYPE html>

```
<html>
<head>
  <title>Pricing Table</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
```

```
<section class="pricing">
```

```
  <div class="card">
    <h2>Basic</h2>
    <h3>$10</h3>
    <ul>
      <li>1 Website</li>
      <li>Basic Support</li>
      <li>1GB Storage</li>
    </ul>
    <button>Buy</button>
  </div>
```

```
  <div class="card">
    <h2>Standard</h2>
    <h3>$20</h3>
    <ul>
      <li>5 Websites</li>
      <li>Priority Support</li>
      <li>10GB Storage</li>
    </ul>
    <button>Buy</button>
  </div>
```

```
  <div class="card">
    <h2>Premium</h2>
    <h3>$30</h3>
    <ul>
      <li>Unlimited Websites</li>
      <li>24/7 Support</li>
      <li>50GB Storage</li>
    </ul>
    <button>Buy</button>
  </div>
```

```
</section>
```

```
</body>
```

```
</html>
```

Css file:

```
body {
  margin: 0;
  font-family: Arial;
}
```

```
.pricing {
  display: flex;
  justify-content: center;
  gap: 20px;
  padding: 40px;
}
```

```

.card {
  background-color: white;
  border: 1px solid gray;
  padding: 20px;
  width: 250px;
  text-align: center;
}

.card:hover {
  transform: scale(1.05);
}

.card ul {
  list-style: none;
  padding: 0;
}

.card ul li {
  margin: 8px 0;
}

button {
  padding: 8px 12px;
  background-color: black;
  color: white;
  border: none;
}

@media (max-width: 768px) {
  .pricing {
    flex-direction: column;
    align-items: center;
  }
}
*****

```

Problem6: Grid Photo Gallery:

Html file:

```

<!DOCTYPE html>
<html>
<head>
  <title>Photo Gallery</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

```

```

<section class="gallery">
  <div class="box">Photo 1</div>
  <div class="box">Photo 2</div>
  <div class="box">Photo 3</div>
  <div class="box">Photo 4</div>
  <div class="box">Photo 5</div>
  <div class="box">Photo 6</div>
  <div class="box">Photo 7</div>
  <div class="box">Photo 8</div>
</section>

```

```

</body>

```

```

</html>
Css file:
body {
    margin: 0;
    font-family: Arial;
}

.gallery {
    display: grid;
    grid-template-columns: repeat(4, 1fr);
    gap: 20px;
    padding: 20px;
}

.box {
    background-color: lightgray;
    padding: 60px;
    text-align: center;
    border-radius: 10px;
    box-shadow: 2px 2px 8px gray;
    transition: background-color 0.3s;
}

.box:hover {
    background-color: lightblue;
}

/* Tablet */
@media (max-width: 992px) {
    .gallery {
        grid-template-columns: repeat(2, 1fr);
    }
}

/* Mobile */
@media (max-width: 576px) {
    .gallery {
        grid-template-columns: repeat(1, 1fr);
    }
}
*****

```

Problem7: Sidebar Layout with Hide/Show on Mobile

Htmlfile:

```

<!DOCTYPE html>
<html>
<head>
    <title>Sidebar Layout</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>

<header class="header">
    <h2>My Website</h2>
</header>

<div class="container">

    <aside class="sidebar">

```

```
    <a href="#">Home</a>
    <a href="#">Shop</a>
    <a href="#">Beauty</a>
    <a href="#">Health</a>
    <a href="#">Support</a>
</aside>

<main class="content">
  <h3>Main Content</h3>
  <p>This is the main content area.</p>
</main>

</div>
```

```
<footer class="footer">
  <p>Footer Section</p>
</footer>
```

```
</body>
</html>
```

Css file:

```
body {
  margin: 0;
  font-family: Arial;
}

.header {
  background-color: black;
  color: white;
  padding: 15px;
  text-align: center;
}

.container {
  display: flex;
}

.sidebar {
  width: 200px;
  background-color: lightgray;
  padding: 20px;
}

.sidebar a {
  display: block;
  text-decoration: none;
  color: black;
  margin: 10px 0;
}

.sidebar a:hover {
  background-color: gray;
  color: white;
}

.content {
  flex: 1;
  padding: 20px;
}
```

```
}

.footer {
  background-color: black;
  color: white;
  text-align: center;
  padding: 10px;
}

/* Mobile */
@media (max-width: 768px) {

  .container {
    flex-direction: column;
  }

  .sidebar {
    width: 100%;
  }

}
```

javaScript problems 18-02-2026

problem 1: ATM Withdrawal Program (if-else + operators)

htmlfile:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>ATM Withdrawal Program</title>
</head>
<body>

  <button onclick="withdraw()">click</button>
  <p id="out"></p>
  <script src="script.js"></script>

</body>
</html>
```

JS file:

```
function withdraw(){
  let balance=Number(prompt(" balance:"))
  let Withdrawal=Number(prompt("how much amount to withdraw:"))

  if (Withdrawal>balance){
    document.getElementById("out").innerHTML="Insufficient Balance";
  }

  else if(Withdrawal<=0){
    document.getElementById("out").innerHTML="Invalid Amount";
  }
  else if(Withdrawal%100==0){
    let remainingBalance=balance-Withdrawal;
    document.getElementById("out").innerHTML="remaining Balance:"+remainingBalance;
  }
  else{
    document.getElementById("out").innerHTML="Enter amount in multiples of 100";
  }
}
*****
```

Problem2: Electricity Bill Calculator

Htmlfile:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Electricity Bill Calculator</title>
</head>
<body>

  <h2>ElectricityBill</h2>

  <button onclick="calculateCurrent()">click</button>
  <p id="out"></p>
  <script src="script.js"></script>
```



```

</body>
</html>
Jss file:
function calculateCurrent(){
  let units=Number(prompt("enter units:"))
  let bill=0;

  if(units<=100){
    bill=units*2;
  }
  else if(101<units<200){
    bill=units*3;
  }
  else if(201<units<300){
    bill=units*5;
  }
  else{
    bill=units*8;
  }
  document.getElementById("out").innerHTML=
  "units consumed: " + units + "<br>" + "Total bill amount:" + bill;

}
*****

```

Problem3: Weekday Finder using Switch

Html file

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Weekday Finder using Switch</title>
</head>
<body>

  <h2>Week-days</h2>

  <button onclick="week()">click</button>
  <p id="out"></p>
  <script src="script.js"></script>

</body>
</html>

```

Js file:

```

function week(){
  let num=Number(prompt("enter number:"))
  let day="";

  switch(num){
    case 1:
      day="Monday";
      break;
    case 2:
      day="Tuesday";
      break;
    case 3:
      day="Wednesday";
      break;
  }
}

```

```

    case 4:
        day="Thrusday";
        break;
    case 5:
        day="Friday";
        break;
    case 6:
        day="Saturday";
        break;
    case 7:
        day="Sunday";
        break;

    default:
        showOut("invalid");

}let type=""
if(num==6 || num==7){
    type="weekend";}

else{

    type="weekday";
}

document.getElementById("out").innerHTML="day is " + day + "<br>" + "type is " + type;
}
*****

```

Problem4: Password Validation Program

Html file:

```

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Weekday Finder using Switch</title>
</head>
<body>

    <h2> Password Validation Program</h2>

    <button onclick="validp()">click</button>
    <p id="out"></p>
    <script src="script.js"></script>

</body>
</html>

```

JS file:

```

function validp() {
    let password = prompt("Enter password:");

    if (password.length < 8) {
        document.getElementById("out").innerHTML =
            "Please enter at least 8 characters<br>Invalid password + ";
    }
    else if (!/[A-Z]/.test(password)) {

```

```

document.getElementById("out").innerHTML =
    "Please enter at least one uppercase letter<br>Invalid password +";
}
else if (!/[0-9]/.test(password)) {
    document.getElementById("out").innerHTML =
        "Please enter at least one digit<br>Invalid password +";
}
else {
    document.getElementById("out").innerHTML
        ="Valid password";
    }
}
}

```

Problem5: Traffic Signal Simulator

Html file:

```

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Traffic Signal Simulator</title>
</head>
<body>

    <h2> Traffic Signal Simulator</h2>

    <button onclick="Trafficcheck()">click</button>
    <p id="out"></p>
    <script>function Trafficcheck() {

        let colour = prompt("Enter the color:");

        switch (colour.toLowerCase()) {

            case "red":
                console.log("Stop");
                break;

            case "yellow":
                console.log("Get Ready");
                break;

            case "green":
                console.log("Go");
                break;

            default:
                console.log("Enter valid input");
        }
    }
</script>

</body>
</html>

```

Problem6: Number Analysis Tool

<!DOCTYPE html>

```

<html>
<head>
  <title>Number Analysis Tool</title>
</head>
<body>

<h2>Number Analysis Tool</h2>
<button onclick="checkNumber()">Click</button>

<p id="out"></p>

<script>
function checkNumber() {

  let num = Number(prompt("Enter a number:"));
  let output = "";

  // Positive or Negative (Ternary Operator)
  let result = (num >= 0) ? "Positive Number" : "Negative Number";
  output += result + "<br>";

  // Even or Odd (if-else) if
  (num % 2 === 0) {
    output += "Even Number<br>";
  } else {
    output += "Odd Number<br>";
  }

  //Loop from 1 to given number
  output += "Numbers from 1 to " + num + " :<br>";

  for (let i = 1; i <= num; i++) {
    output += i + " ";
  }

  document.getElementById("out").innerHTML = output;
}
</script>

</body>
</html>

```

Problem7: Dynamic Student Profile Manager

```

<!DOCTYPE html>
<html>
<head>
  <title>Student Profile Manager</title>
  <style>
    #profile {
      border: 2px solid black;
      padding: 10px;
      width: 250px;
      margin-top: 10px;
    }
  </style>
</head>
<body>

```

```

<h2>Student Profile Manager</h2>

<button onclick="updateStudentProfile(student)">Show Profile</button>
<button onclick="updateMarks()">Update Marks</button>

<div id="profile"></div>

<script>

let student = {
  name: "Rahul",
  rollNo: 101,
  marks: 85
};

function updateStudentProfile(studentObj) {

  document.getElementById("profile").innerHTML =
    "<b>Name:</b> " + studentObj.name + "<br>" +
    "<b>Roll No:</b> " + studentObj.rollNo + "<br>" +
    "<b>Marks:</b> " + studentObj.marks;
}

function updateMarks() {

  let newMarks = prompt("Enter new marks:");

  student.marks = newMarks;

  updateStudentProfile(student);
}

</script>

</body>
</html>

```

```

*****

Part2
Problem:1Discount Calculator (Function + Parameters + DOM)
<!DOCTYPE html>
<html>
<head>
  <title>Discount Calculator</title>
</head>
<body>

<h2>Discount Calculator</h2>

<label>Product Price:</label>
<input type="number" id="price"><br><br>

<label>Discount Percentage:</label>
<input type="number" id="discount"><br><br>

<button onclick="showResult()">Calculate</button>

<p id="result"></p>

<script>

```

```

function calculateDiscount(price, discount) {

    let finalPrice = price - (price * discount / 100);
    return finalPrice;
}

function showResult() {

    let price = Number(document.getElementById("price").value);
    let discount = Number(document.getElementById("discount").value);

    let finalAmount = calculateDiscount(price, discount);

    document.getElementById("result").innerHTML =
        "Final Price after discount: " + finalAmount;
}

```

```

</script>

```

```

</body>
</html>

```

```

*****

```

Problem8: Filter Passing Marks (filter() + DOM)

```

<!DOCTYPE html>

```

```

<html>

```

```

<head>

```

```

    <title>Filter Passing Marks</title>

```

```

</head>

```

```

<body>

```

```

<h2>Filter Passing Marks</h2>

```

```

<button onclick="showMarks()">Show Result</button>

```

```

<p id="output"></p>

```

```

<script>

```

```

let marks = [35, 70, 88, 45, 92, 60, 25];

```

```

function showMarks() {

```

```

    let passed = marks.filter(function(mark) {
        return mark >= 50;
    });

```

```

    document.getElementById("output").innerHTML =
        "Original Marks: " + marks + "<br><br>" +
        "Passed Marks: " + passed + "<br><br>" +
        "Total Passed Students: " + passed.length;
}

```

```

</script>

```

```

</body>

```

```

</html>

```

```

*****

```

Problem9: Employee Object System (Objects + Functions)

```
<!DOCTYPE html>
<html>
<head>
  <title>Employee Salary Update</title>
</head>
<body>

<h2>Employee Salary Manager</h2>

<button onclick="showEmployee()">Show Employee</button>
<button onclick="updateSalary()">Increase Salary</button>

<p id="output"></p>

<script>

let employee = {
  name: "Amit",
  salary: 40000,
  department: "IT"
};

function increaseSalary(emp, percentage) {

  emp.salary = emp.salary + (emp.salary * percentage / 100);
  return emp.salary;
}

function showEmployee() {

  document.getElementById("output").innerHTML =
    "Name: " + employee.name + "<br>" +
    "Salary: " + employee.salary + "<br>" +
    "Department: " + employee.department;
}

function updateSalary() {

  let percent = Number(prompt("Enter increase percentage:"));

  let newSalary = increaseSalary(employee, percent);

  let level;

  if (newSalary > 50000) {
    level = "Senior Employee";
  } else {
    level = "Junior Employee";
  }

  document.getElementById("output").innerHTML =
    "Name: " + employee.name + "<br>" +
    "Salary: " + newSalary + "<br>" +
    "Department: " + employee.department + "<br>" +
    level;
}
```

```

</script>

</body>
</html>
*****

Problem10: Scope Debug Program (Global vs Local vs Block)
<!DOCTYPE html>
<html>
<head>
  <title>Scope Debug Program</title>
</head>
<body>

<h2>Scope Debug Program</h2>

<button onclick="checkScope()">Run</button>

<p id="output"></p>

<script>

let globalVar = "I am Global";

function checkScope() {

  let functionVar = "I am Function";

  let text = "";

  {
    let blockVar = "I am Block";
    console.log(globalVar);
    console.log(functionVar);
    console.log(blockVar);

    text += globalVar + "<br>";
    text += functionVar + "<br>";
    text += blockVar + "<br>";
  }
  // text += blockVar; it stops execution as it is not accessing
  document.getElementById("output").innerHTML = text;

  console.log(blockVar);
  text += blockVar;
}

</script>

</body>
</html>
*****

Problem11: DOM Color Theme Changer (Events + DOM)
<!DOCTYPE html>
<html>
<head>
  <title>DOM Color Theme Changer</title>
</head>
<body>

```



```
<h2 id="heading">DOM Color Theme Changer</h2>
<p> another para</p>
```

```
<button onclick="setTheme('dark')">Dark Mode</button>
<button onclick="setTheme('light')">Light Mode</button>
<button onclick="setTheme('blue')">Blue Mode</button>
```

```
<br><br>
```

```
<p id="modeMsg">hello it will change color DOM manipulation </p>
```

```
<script>
```

```
function setTheme(mode) {
```

```
    let msg = document.getElementById("modeMsg");
```

```
    if (mode === "dark") {
```

```
        document.body.style.backgroundColor = "black";
        document.body.style.color = "pink";
        document.getElementById("heading").style.color = "yellow";
```

```
    }
```

```
    else if (mode === "blue") {
```

```
        document.body.style.backgroundColor = "lightblue";
        document.body.style.color = "black";
        document.getElementById("heading").style.color = "yellow";
```

```
    }
```

```
    else {
```

```
        document.body.style.backgroundColor = "white";
        document.body.style.color = "black";
        document.getElementById("heading").style.color = "black";
```

```
    }
```

```
}
```

```
</script>
```

```
</body>
```

```
</html>
```

```
*****
```

Problem12: Search Numbers Greater than Input (Function + filter + User Input)

```
let numbers = [10, 25, 60, 75, 90, 15, 45, 120];
```

```
function filterNumbers() {
```

```
    let inputValue = document.getElementById("userInput").value;
```

```
    inputValue = Number(inputValue); // convert string to number
```

```
    let filtered = numbers.filter(function(num) {
```

```
        return num > inputValue;
```

```
    });
```

```

// Extra Challenge: sort ascending
filtered.sort(function(a, b) {
    return a - b;
});

document.getElementById("result").innerHTML =
    "Numbers greater than " + inputValue + ": " + filtered;
}

*****

Date:19-02-2026
Probleme1: Location-Based Weather Logger
<h2>Location Based Weather Logger</h2>

<button onclick="getMyLocation()">Get My Location</button>

<p id="locationOutput"></p>

<h3>Location History</h3>
<ul id="historyList"></ul>

<script>

function getMyLocation() {

    let output = document.getElementById("locationOutput");

    if (navigator.geolocation) {
        navigator.geolocation.getCurrentPosition(showPosition, showError);
    } else {
        output.innerHTML = "Geolocation is not supported by your browser.";
    }
}

function showPosition(position) {

    let latitude = position.coords.latitude;
    let longitude = position.coords.longitude;

    document.getElementById("locationOutput").innerHTML =
        "Latitude: " + latitude +
        "<br>Longitude: " + longitude;

    saveLocation(latitude, longitude);
}

function showError(error) {

    let output = document.getElementById("locationOutput");

    if (error.code === 1) {
        output.innerHTML = "Permission Denied! Please allow location access.";
    }
    else if (error.code === 2) {
        output.innerHTML = "Location not available.";
    }
    else if (error.code === 3) {
        output.innerHTML = "Request timed out.";
    }
}

```

```

    }
    else {
        output.innerHTML = "Unknown error occurred.";
    }
}
function saveLocation(lat, lon) {

    let storedData = localStorage.getItem("locationHistory");

    if (storedData) {
        storedData = JSON.parse(storedData);
    } else {
        storedData = [];
    }

    let locationObject = {
        latitude: lat,
        longitude: lon,
        time: new Date().toLocaleString()
    };

    storedData.unshift(locationObject);

    if (storedData.length > 5) {
        storedData.pop();
    }

    localStorage.setItem("locationHistory", JSON.stringify(storedData));

    document.getElementById("locationOutput").innerHTML +=
        "<br>Location saved in localStorage.";
}
window.onload = function() {

    let savedLocations = localStorage.getItem("locationHistory");

    if (savedLocations) {

        savedLocations = JSON.parse(savedLocations);

        let historyList = document.getElementById("historyList");
        historyList.innerHTML = "";

        for (let i = 0; i < savedLocations.length; i++) {

            let li = document.createElement("li");

            li.innerHTML =
                "Latitude: " + savedLocations[i].latitude +
                " Longitude: " + savedLocations[i].longitude +
                " Time: " + savedLocations[i].time;

            historyList.appendChild(li);
        }
    }
}

```

//when i check my location i got Location not available

```

</script>
</body>
</html>
*****

Probleme2: Mini Expense Tracker using Client-Side Database(indexDB)
Html file: <!DOCTYPE html>
<html>
<head>
  <title>Mini Expense Tracker</title>
</head>
<body>

<h1>Mini Expense Tracker</h1>

<h2>Add Expense</h2>
<input type="text" id="title" placeholder="Expense Title">
<input type="number" id="amount" placeholder="Amount">
<input type="date" id="date">
<button onclick="addExpense()">Add Expense</button>
<button onclick="showExpenses()">View Expenses</button>

<h2>Delete Expense</h2>
<input type="number" id="deleteId" placeholder="Enter ID">
<button onclick="deleteExpense()">Delete Expense</button>

<div id="output">
<b>Output will appear here...</b>
</div>

<script src="script.js"></script>

</body>
</html>
Js file:
let db;

// OPEN / CREATE DATABASE
let request = indexedDB.open("ExpenseDatabase", 1);

request.onupgradeneeded = function (event) {

  db = event.target.result;

  let objectStore = db.createObjectStore("expenses", { keyPath: "id", autoIncrement: true });

  objectStore.createIndex("title", "title", { unique: false });
  objectStore.createIndex("amount", "amount", { unique: false });
  objectStore.createIndex("date", "date", { unique: false });

  console.log("Database created successfully!");
};

request.onsuccess = function (event) {
  db = event.target.result;
  console.log("Database opened successfully!");
};

```

```

request.onerror = function () {
    document.getElementById("output").innerHTML = "Error opening database!";
};

// ADD EXPENSE
function addExpense() {

    let title = document.getElementById("title").value;
    let amount = document.getElementById("amount").value;
    let date = document.getElementById("date").value;

    if (title === "" || amount === "" || date === "") {
        document.getElementById("output").innerHTML = "Please fill all fields!";
        return;
    }

    let transaction = db.transaction(["expenses"], "readwrite");

    transaction.onerror = function () {
        document.getElementById("output").innerHTML = "Transaction error!";
    };

    let objectStore = transaction.objectStore("expenses");

    let expenseData = {
        title: title,
        amount: Number(amount),
        date: date
    };

    let request = objectStore.add(expenseData);

    request.onsuccess = function () {
        document.getElementById("output").innerHTML = "Expense added successfully!";
        document.getElementById("title").value = "";
        document.getElementById("amount").value = "";
        document.getElementById("date").value = "";
    };

    request.onerror = function () {
        document.getElementById("output").innerHTML = "Error adding expense!";
    };
}

// VIEW EXPENSES
function showExpenses() {

    let transaction = db.transaction(["expenses"], "readonly");

    transaction.onerror = function () {
        document.getElementById("output").innerHTML = "Transaction error!";
    };

    let objectStore = transaction.objectStore("expenses");

    let request = objectStore.getAll();

    request.onsuccess = function (event) {

```

```

    let expenses = event.target.result;

    if (expenses.length === 0) {
        document.getElementById("output").innerHTML = "No expenses found!";
        return;
    }

    let output = "<h3>Expense List:</h3>";

    expenses.forEach(function (expense) {
        output += "ID: " + expense.id +
            " | Title: " + expense.title +
            " | Amount: " + expense.amount +
            " | Date: " + expense.date + "<br>";
    });

    document.getElementById("output").innerHTML = output;
};

request.onerror = function () {
    document.getElementById("output").innerHTML = "Error fetching expenses!";
};
}

// DELETE EXPENSE
function deleteExpense() {

    let id = Number(document.getElementById("deleteld").value);

    if (!id) {
        document.getElementById("output").innerHTML = "Enter valid ID!";
        return;
    }

    let transaction = db.transaction(["expenses"], "readwrite");

    transaction.onerror = function () {
        document.getElementById("output").innerHTML = "Transaction error!";
    };

    let objectStore = transaction.objectStore("expenses");

    let request = objectStore.delete(id);

    request.onsuccess = function () {
        document.getElementById("output").innerHTML = "Expense deleted successfully!";
        document.getElementById("deleteld").value = "";
    };

    request.onerror = function () {
        document.getElementById("output").innerHTML = "Error deleting expense!";
    };
}
// used indexDB
*****

```

problem:3 (Attendance Tracker System)

Htmlfile: <!DOCTYPE html>

```

<html>
<head>
  <title>Attendance Tracker</title>
</head>
<body>

<h1>Attendance Tracker System</h1>

<h2>Mark Attendance</h2>

<input type="text" id="studentName" placeholder="Student Name">
<input type="date" id="attendanceDate">

<select id="status">
  <option value="Present">Present</option>
  <option value="Absent">Absent</option>
</select>

<br><br>

<button onclick="saveAttendance()">Save Attendance</button>
<button onclick="showAttendance()">Show Attendance</button>
<button onclick="clearAttendance()">Clear Attendance Database</button>

<p id="message"></p>

<div id="output"></div>

<script src="script.js"></script>

</body>
</html>

```

Js file:

```
let db;
```

```
// OPEN / CREATE DATABASE
```

```
let request = indexedDB.open("AttendanceDatabase", 1);
```

```
request.onupgradeneeded = function (event) {
```

```
  db = event.target.result;
```

```
  let objectStore = db.createObjectStore("attendance", {
    keyPath: "id",
    autoIncrement: true
  });
```

```
  objectStore.createIndex("name", "name", { unique: false });
  objectStore.createIndex("date", "date", { unique: false });
  objectStore.createIndex("status", "status", { unique: false });
```

```
  console.log("Database created successfully!");
};
```

```
request.onsuccess = function (event) {
```

```
  db = event.target.result;
```

```
  console.log("Database opened successfully!");
```

```
};
```

```
request.onerror = function () {  
    document.getElementById("message").innerHTML = "Error opening database!";  
};
```

```
// SAVE ATTENDANCE
```

```
function saveAttendance() {
```

```
    let name = document.getElementById("studentName").value;  
    let date = document.getElementById("attendanceDate").value;  
    let status = document.getElementById("status").value;
```

```
    if (name === "" || date === "") {  
        document.getElementById("message").innerHTML =  
            "Please enter student name and date!";  
        return;  
    }
```

```
    let transaction = db.transaction(["attendance"], "readwrite");
```

```
    transaction.onerror = function () {  
        document.getElementById("message").innerHTML =  
            "Transaction error!";  
    };  
};
```

```
let objectStore = transaction.objectStore("attendance");
```

```
let attendanceData = {  
    name: name,  
    date: date,  
    status: status  
};
```

```
let request = objectStore.add(attendanceData);
```

```
request.onsuccess = function () {
```

```
    document.getElementById("message").innerHTML =  
        "Attendance saved successfully!";
```

```
    // Store last student name in sessionStorage  
    sessionStorage.setItem("lastStudent", name);
```

```
    document.getElementById("studentName").value = "";  
    document.getElementById("attendanceDate").value = "";  
};
```

```
request.onerror = function () {  
    document.getElementById("message").innerHTML =  
        "Error saving attendance!";  
};  
}
```

```
// SHOW ATTENDANCE
```

```
function showAttendance() {
```

```
    let transaction = db.transaction(["attendance"], "readonly");
```



```

transaction.onerror = function () {
    document.getElementById("message").innerHTML =
        "Transaction error!";
};

let objectStore = transaction.objectStore("attendance");

let request = objectStore.getAll();

request.onsuccess = function (event) {

    let records = event.target.result;

    if (records.length === 0) {
        document.getElementById("output").innerHTML =
            "No attendance records found!";
        return;
    }

    let table = "<table border='1'>";
    table += "<tr><th>ID</th><th>Name</th><th>Date</th><th>Status</th></tr>";

    records.forEach(function (record) {
        table += "<tr>";
        table += "<td>" + record.id + "</td>";
        table += "<td>" + record.name + "</td>";
        table += "<td>" + record.date + "</td>";
        table += "<td>" + record.status + "</td>";
        table += "</tr>";
    });

    table += "</table>";

    document.getElementById("output").innerHTML = table;
};

request.onerror = function () {
    document.getElementById("message").innerHTML =
        "Error fetching attendance!";
};
}

```

// CLEAR DATABASE

```

function clearAttendance() {

    let transaction = db.transaction(["attendance"], "readwrite");

    transaction.onerror = function () {
        document.getElementById("message").innerHTML =
            "Transaction error!";
    };

    let objectStore = transaction.objectStore("attendance");

    let request = objectStore.clear();

    request.onsuccess = function () {
        document.getElementById("output").innerHTML =

```

```

        "All attendance records cleared!";
    };

    request.onerror = function () {
        document.getElementById("message").innerHTML =
            "Error clearing database!";
    };
}
*****

```

Probleme4: **Delivery Location Saver**

Html file:

```
<h2>Delivery Location Saver</h2>
```

```
<button onclick="detectLocation()">Detect My Location</button>
```

```
<button onclick="viewLocationHistory()">View Location History</button>
```

```
<p id="locationOutput"></p>
```

```
<div id="historyOutput"></div>
```

```
<script src="script.js"></script>
```

Js file:

```
let db;
```

```
// OPEN / CREATE DATABASE
```

```
let request = indexedDB.open("DeliveryDB", 1);
```

```
request.onupgradeneeded = function (event) {
```

```
    db = event.target.result;
```

```
    let objectStore = db.createObjectStore("deliveryLocations", {
        keyPath: "id",
        autoIncrement: true
    });
```

```
    objectStore.createIndex("latitude", "latitude", { unique: false });
    objectStore.createIndex("longitude", "longitude", { unique: false });
    objectStore.createIndex("dateTime", "dateTime", { unique: false });
```

```
    console.log("Database created successfully!");
};
```

```
request.onsuccess = function (event) {
    db = event.target.result;
    console.log("Database opened successfully!");
};
```

```
request.onerror = function () {
    document.getElementById("locationOutput").innerHTML =
        "Error opening database!";
};
```

```
// DETECT LOCATION
```

```
function detectLocation() {
```

```
    if (navigator.geolocation) {
```

```

        navigator.geolocation.getCurrentPosition(showPosition, showError);

    } else {

        document.getElementById("locationOutput").innerHTML =
            "Geolocation not supported.";
    }
}

// SUCCESS CALLBACK
function showPosition(position) {

    let latitude = position.coords.latitude;
    let longitude = position.coords.longitude;

    document.getElementById("locationOutput").innerHTML =
        "Latitude: " + latitude +
        "<br>Longitude: " + longitude;

    // STORE LATEST IN LOCALSTORAGE
    let lastLocation = {
        latitude: latitude,
        longitude: longitude
    };

    localStorage.setItem("lastLocation",
        JSON.stringify(lastLocation));

    // STORE IN INDEXEDDB
    let transaction = db.transaction(["deliveryLocations"], "readwrite");

    transaction.onerror = function () {
        document.getElementById("locationOutput").innerHTML =
            "Transaction error!";
    };

    let objectStore = transaction.objectStore("deliveryLocations");

    let locationData = {
        latitude: latitude,
        longitude: longitude,
        dateTime: new Date().toLocaleString()
    };

    let request = objectStore.add(locationData);

    request.onsuccess = function () {
        console.log("Location stored in IndexedDB");
    };

    request.onerror = function () {
        document.getElementById("locationOutput").innerHTML =
            "Error saving location!";
    };
}

// ERROR CALLBACK
function showError(error) {

```

```

if (error.code === 1) {
    document.getElementById("locationOutput").innerHTML =
        "Location access denied";
}
else {
    document.getElementById("locationOutput").innerHTML =
        "Error fetching location";
}
}

```

// VIEW LOCATION HISTORY

```

function viewLocationHistory() {

    let transaction = db.transaction(["deliveryLocations"], "readonly");

    transaction.onerror = function () {
        document.getElementById("historyOutput").innerHTML =
            "Transaction error!";
    };

    let objectStore = transaction.objectStore("deliveryLocations");

    let request = objectStore.getAll();

    request.onsuccess = function (event) {

        let locations = event.target.result;

        if (locations.length === 0) {
            document.getElementById("historyOutput").innerHTML =
                "No location history found!";
            return;
        }

        let output = "<h3>Location History:</h3>";

        locations.forEach(function (loc) {
            output +=
                "ID: " + loc.id +
                " | Latitude: " + loc.latitude +
                " | Longitude: " + loc.longitude +
                " | DateTime: " + loc.dateTime +
                "<br>";
        });

        document.getElementById("historyOutput").innerHTML = output;
    };

    request.onerror = function () {
        document.getElementById("historyOutput").innerHTML =
            "Error retrieving history!";
    };
}
// getting two errors
Error fetching location

```

//No location history found!

2nd part

Problem:1 Student Marks Analyzer

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Student Marks Analyzer</title>
<style>
body {
  font-family: Arial;
  padding: 20px;
}
.box {
  padding: 10px;
  border: 1px solid black;
  margin-top: 15px;
}
</style>
</head>
<body>
```

```
<h1>Student Marks Analyzer</h1>
<h3>Analyze Marks</h3>
```

```
<div class="box" id="output">
Output will appear here...
</div>
```

```
<script type="module" src="main.js"></script>
```

```
</body>
</html>
```

```
Main.js: import { marks } from "./marks.js";
```

```
const output = document.getElementById("output");
```

```
// Display marks using map()
const marksList = marks.map(mark => `${mark} `).join(",");
```

```
// Calculate total using reduce()
const total = marks.reduce((sum, mark) => sum + mark, 0);
```

```
// Calculate average
const average = total / marks.length;
```

```
// Pass / Fail
const result = average >= 50 ? "Pass" : "Fail";
```

```
// Display output
output.innerHTML = `
  <h3>Marks Analysis</h3>
  Marks: ${marksList} <br>
  Total: ${total} <br>
  Average: ${average.toFixed(2)} <br>
  Result: <b>${result}</b>
`;
Marks.js
```

```
export const marks = [85, 72, 90, 60, 40, 88];
*****
```

Problem2: Student Result Analyzer

Html file:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Student Result Analyzer</title>
<style>
body {
  font-family: Arial;
  padding: 20px;
}
button {
  padding: 10px;
  margin: 5px;
  cursor: pointer;
}
.box {
  padding: 10px;
  border: 1px solid black;
  margin-top: 15px;
}
.student {
  border: 1px solid gray;
  padding: 8px;
  margin: 5px 0;
}
</style>
</head>
<body>

<h1>Student Result Analyzer</h1>

<button onclick="sortStudents()">Sort Students by Marks</button>
<button onclick="filterPassed()">Filter Passed Students</button>
<button onclick="calculateAverage()">Calculate Average Marks</button>
<button onclick="displayStudents()">Display All Students</button>

<div class="box" id="output">
Output will appear here...
</div>

<script type="module" src="main.js"></script>

</body>
</html>
```

Students.js:

```
export const students = [
  { name: "Kunal", marks: 85 },
  { name: "Rahul", marks: 40 },
  { name: "Neha", marks: 72 },
  { name: "Amit", marks: 55 },
  { name: "Priya", marks: 30 }
];
```

Main.js:

```

import { students } from "./students.js";

const output = document.getElementById("output");

// A) SORT STUDENTS BY MARKS
window.sortStudents = () => {

    const sorted = [...students].sort((a, b) => b.marks - a.marks);

    displayList(sorted, "Students Sorted by Marks (High → Low)");
};

// B) FILTER PASSED STUDENTS
window.filterPassed = () => {

    const passed = students.filter(student => student.marks >= 50);

    displayList(passed, "Passed Students (Marks ≥ 50)");
};
// C) CALCULATE AVERAGE MARKS

window.calculateAverage = () => {

    const total = students.reduce((sum, student) => sum + student.marks, 0);

    const average = total / students.length;

    output.innerHTML = `
        <h3>Average Calculation</h3>
        Total Marks: ${total} <br>
        Average Marks: ${average.toFixed(2)}
    `;
};
// D) DISPLAY STUDENTS DYNAMICALLY
window.displayStudents = () => {

    displayList(students, "All Students");
};

// COMMON DISPLAY FUNCTION
const displayList = (list, title) => {

    const html = list.map(student => {

        const grade =
            student.marks >= 80 ? "A" :
            student.marks >= 60 ? "B" :
            student.marks >= 50 ? "C" :
            "Fail";

        return `
            <div class="student">
                <b>${student.name}</b><br>
                Marks: ${student.marks}<br>
                Grade: ${grade}
            </div>
        `;
    });
    return html.join("");
};

```

```

    output.innerHTML = `
      <h3>${title}</h3>
      ${html}
    `;
  };
}
*****

```

Probleme3: Async Order Status Tracker

Html file:

```

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Async Order Status Tracker</title>
<style>
body {
  font-family: Arial;
  padding: 20px;
}
button {
  padding: 10px;
  margin: 5px;
  cursor: pointer;
}
.box {
  padding: 10px;
  border: 1px solid black;
  margin-top: 15px;
}
</style>
</head>
<body>

<h1>Async Order Status Tracker</h1>

<button onclick="callbackVersion()">Track Order using Callback</button>
<button onclick="promiseVersion()">Track Order using Promise</button>
<button onclick="asyncAwaitVersion()">Track Order using Async/Await</button>

<div class="box" id="output">
Output will appear here...
</div>

<script type="module" src="main.js"></script>

</body>
</html>

```

Orders.js:

```

export const orders = [
  { orderId: 101, status: "Placed" },
  { orderId: 102, status: "Shipped" },
  { orderId: 103, status: "Delivered" }
];

```

Main.js:

```

import { orders } from "./orders.js";

const output = document.getElementById("output");

```



```
// A) CALLBACK VERSION
```

```
function trackOrderCallback(orderId, callback) {

  setTimeout(() => {

    const order = orders.find(o => o.orderId === orderId);

    if (order) {
      callback(`Order ID: ${orderId} <br>Status: ${order.status}`);
    } else {
      callback("Order not found!");
    }

  }, 2000);
}
```

```
window.callbackVersion = () => {

  const id = parseInt(prompt("Enter Order ID:"));

  output.innerHTML = "Tracking order using Callback...";

  trackOrderCallback(id, (message) => {
    output.innerHTML = message;
  });
};
```

```
// B) PROMISE VERSION
```

```
function trackOrderPromise(orderId) {

  return new Promise((resolve, reject) => {

    setTimeout(() => {

      const order = orders.find(o => o.orderId === orderId);

      if (order) {
        resolve(`Order ID: ${orderId} <br>Status: ${order.status}`);
      } else {
        reject("Order not found!");
      }

    }, 2000);

  });
}
```

```
window.promiseVersion = () => {

  const id = parseInt(prompt("Enter Order ID:"));

  output.innerHTML = "Tracking order using Promise...";

  trackOrderPromise(id)
    .then(result => {
```

```

        output.innerHTML = result;
    })
    .catch(error => {
        output.innerHTML = error;
    });
};

```

// C) ASYNC / AWAIT VERSION

```

async function trackOrderAsync(orderId) {

```

```

    const result = await trackOrderPromise(orderId);
    return result;
}

```

```

window.asyncAwaitVersion = async () => {

```

```

    const id = parseInt(prompt("Enter Order ID:"));

```

```

    output.innerHTML = "Tracking order using Async/Await...";

```

```

    try {
        const result = await trackOrderAsync(id);
        output.innerHTML = result;
    } catch (error) {
        output.innerHTML = error;
    }
}

```

```

};
*****

```

Date: 20-02-2026(day8)

Problem1: Advanced Object Destructuring

Js file:

```

let student = {
    id: 101,
    name: "sony",
    age: 26,
    course: "Full Stack",
    address: {
        city: "Bangalore",
        state: "Karnataka"
    }
};

```

```

let {
    name: studentName,
    course,
    address: { city },
    country = "India"
} = student;

```

```

console.log("Student: " + studentName);

```

```

console.log("Course: " + course);

```

```

console.log("City: " + city);

```

```

*****

```

Problem 2: Function with Multiple Default Parameters

```

function calculateOrder(price, quantity = 1, discount = 0) {

```

```

    if (price === undefined) {

```

```

        console.log("Error: Price is required!");
        return;
    }

    let total = price * quantity;
    let discountAmount = total * discount / 100;
    let finalAmount = total - discountAmount;

    console.log(`Price: ${price}`);
    console.log(`Quantity: ${quantity}`);
    console.log(`Discount: ${discount}%`);
    console.log(`Final Amount: ${finalAmount}`);
}

// Example Calls
calculateOrder(200);
calculateOrder(200, 3);
calculateOrder(200, 3, 10);
calculateOrder();
*****

```

Problem 3: Improve Calculator

Html file:

```

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Simple Calculator</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<div class="calculator">
    <h2>Calculator</h2>
    <input type="text" id="result" readonly>
    <div class="buttons">
        <button onclick="clearResult()">C</button>
        <button onclick="deleteLast()">DEL</button>
        <button onclick="appendValue('/')">/</button>
        <button onclick="appendValue('*')">*</button>
        <button onclick="appendValue('7')">7</button>
        <button onclick="appendValue('8')">8</button>
        <button onclick="appendValue('9')">9</button>
        <button onclick="appendValue('.')">.</button>
        <button onclick="appendValue('4')">4</button>
        <button onclick="appendValue('5')">5</button>
        <button onclick="appendValue('6')">6</button>
        <button onclick="appendValue('+')">+</button>
        <button onclick="appendValue('1')">1</button>
        <button onclick="appendValue('2')">2</button>
        <button onclick="appendValue('3')">3</button>
        <button onclick="calculate()">=</button>
        <button onclick="appendValue('0')" class="zero">0</button>
        <button onclick="appendValue('.')">.</button>
    </div>
</div>
<script src="script.js"></script>
</body>
</html>

```

Css file:

```
body {
  font-family: Arial;
  background-color: #f4f4f4;
  display: flex;
  justify-content: center;
  align-items: center;
  height: 100vh;
}

.calculator {
  background-color: white;
  padding: 20px;
  border-radius: 10px;
  width: 300px;
  box-shadow: 0px 0px 10px gray;
  text-align: center;
}

#result {
  width: 100%;
  height: 50px;
  font-size: 22px;
  margin-bottom: 15px;
  padding: 10px;
  text-align: right;
}

.buttons {
  display: grid;
  grid-template-columns: repeat(4, 1fr);
  gap: 10px;
}

button {
  padding: 15px;
  font-size: 18px;
  cursor: pointer;
  border: none;
  background-color: lightgray;
  border-radius: 5px;
}

button:hover {
  background-color: gray;
  color: white;
}

.zero {
  grid-column: span 2;
}
```

Js file:

```
let resultBox = document.getElementById("result");
```

```
function appendValue(value) {
```

```
  let current = resultBox.value;
  let operators = ["+", "-", "*", "/"];
  let lastChar = current.slice(-1);
```

```
  // Prevent two operators together
```

```

if (operators.includes(lastChar) && operators.includes(value)) {
    return;
}

// Prevent multiple dots in one number
if (value === ".") {

    let parts = current.split(/[\+\-\*\V]/);
    let lastNumber = parts[parts.length - 1];

    if (lastNumber.includes(".")) {
        return;
    }

    // Prevent dot immediately after operator
    if (operators.includes(lastChar)) {
        resultBox.value += "0.";
        return;
    }
}

resultBox.value += value;
}

function clearResult() {
    resultBox.value = "";
}

function deleteLast() {
    resultBox.value = resultBox.value.slice(0, -1);
}

function calculate() {

    let expression = resultBox.value;
    let operators = ["+", "-", "*", "/"];
    let lastChar = expression.slice(-1);

    // Alert if empty
    if (expression === "") {
        alert("Please enter a value first!");
        return;
    }

    // Prevent calculation if expression ends with operator
    if (operators.includes(lastChar)) {
        alert("Expression cannot end with operator!");
        return;
    }

    try {
        // Safer than eval()
        let answer = Function("return " + expression)();
        resultBox.value = answer;
    }
    catch (error) {
        resultBox.value = "Error";
    }
}

```

```

}
*****

Probleme4: Mini Project: Object + DOM + Default Params
Html file:
<!DOCTYPE html>
<html>
<head>
  <title>User Greeting Card</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <div class="card">
    <h2>User Greeting Card</h2>

    <input type="text" id="name" placeholder="Enter Name">
    <input type="number" id="age" placeholder="Enter Age">
    <input type="text" id="city" placeholder="Enter City">

    <button onclick="generateCard()">Generate Card</button>

    <div id="output"></div>
  </div>

  <script src="script.js"></script>
</body>
</html>
Css file:
body {
  font-family: Arial;
  background-color: #f4f4f4;
  display: flex;
  justify-content: center;
  align-items: center;
  height: 100vh;
}

.card {
  background: white;
  padding: 20px;
  width: 300px;
  border-radius: 10px;
  box-shadow: 0 0 10px gray;
  text-align: center;
}

input {
  width: 100%;
  padding: 8px;
  margin: 8px 0.01px;
}

button {
  padding: 10px;
  width: 100%;
  background: lightgray;
  border: none;

```

```
    cursor: pointer;
}
```

```
button: hover {
    background: gray;
    color: white;
}
```

```
#output {
    margin-top: 15px;
}
```

Js file:

```
function generateCard() {
```

```
    // Create object from form values
```

```
    let user = {
        name: document.getElementById("name").value,
        age: document.getElementById("age").value,
        city: document.getElementById("city").value
    };

```

```
    showCard(user);
```

```
}
```

```
// Use destructuring + default value
```

```
function showCard({ name, age, city = "India" }) {
```

```
    let outputDiv = document.getElementById("output");
```

```
    // Extra Challenge
```

```
    let status;
```

```
    if (age < 18) {
        status = "Minor";
    } else {
        status = "Adult";
    }

```

```
    outputDiv.innerHTML =
```

```
        "Welcome " + name + "!<br>" +
```

```
        "Age: " + age + "<br>" +
```

```
        "Location: " + city + "<br>" +
```

```
        "Status: " + status;
```

```
}
```